

■■ BondRisk AI: Varthana Credit Analysis

Project Overview & Technical Documentation

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1. PROJECT OVERVIEW

BondRisk AI is a comprehensive **Bond Risk Assessment platform** that automatically extracts financial metrics from OCR-processed documents and calculates credit risk breach probabilities using artificial intelligence and machine learning algorithms. The system is specifically designed to analyze Varthana (an NBFC) financial documents and assess compliance with regulatory covenants.

2. PROJECT ARCHITECTURE

Data Flow:

OCR Documents → Ingestion (DB) → Chunking → Retrieval → LLM Extraction → Risk Engine → Dashboard

3. FILE STRUCTURE & COMPONENTS

The project is organized into three main components:

Component	Location	Purpose
Data Ingestion	src/ingestion/	Process OCR files and store in database
Risk Engine	src/models/	Calculate breach probability using ML
Web Dashboard	src/web_app/	Visualize metrics and risk scores

4. DATA INGESTION PIPELINE

4.1 watcher.py - File System Monitor

Purpose: Monitors the data/ocr_output/varthana/ directory for new OCR files using the watchdog library. When a .txt file is detected, it is automatically classified based on filename (Quarterly_Report or IM) and triggers the ingestion process.

4.2 db_ingestion.py - Database Ingestion & Chunking

Purpose: Processes OCR text and stores it in PostgreSQL (Neon DB). The script:

- Connects to Neon PostgreSQL database
- Creates/retrieves issuer (Varthana) records
- Registers documents with metadata
- Chunks text using RecursiveCharacterTextSplitter (1000 char chunks, 200 char overlap)
- Bulk inserts chunks into database for high performance

4.3 auto_extractor.py - AI-Powered Metric Extraction

Purpose: Uses Google Gemini 2.5 Flash LLM to extract key financial metrics from document chunks. Extracts 4 key metrics: Capital Adequacy Ratio (CRAR), Leverage Ratio, Asset Quality (Net NPA), and Security Cover. Evaluates compliance status and stores results in database.

4.4 verify.py - Data Verification Tool

Purpose: Validates data retrieval from database by searching for keywords (e.g., 'Leverage') and returning top matching chunks. Used for testing and debugging the retrieval pipeline.

Database Schema:

issuers → documents → document_sections → financial_metrics

5. RISK CALCULATION ENGINE

5.1 risk_engine.py - ML-Based Risk Scoring

Purpose: Calculates bond breach probability using XGBoost machine learning model. The engine:

- Fetches latest 4 financial metrics from database
- Cleans data (converts percentages/ratios to floats)
- Calculates safety headroom margins
- Runs XGBoost inference to predict breach probability
- Outputs risk score as 0-100%
- Uses weighted heuristic fallback if model unavailable

Risk Scoring Formula:

Probability = (1 - CAR_headroom) × 0.4 + (1 - Leverage_headroom) × 0.6

6. WEB DASHBOARD

6.1 app.py - Streamlit Interactive Dashboard

Features:

- Page Title: BondRisk AI: Varthana Credit Analysis
- Left Column: Financial metrics table + risk insights
- Right Column: Risk probability gauge chart
- Color coding: Green (0-30%), Orange (30-70%), Red (70-100%)

Technology Stack: Streamlit (UI), Plotly (visualization)

7. COMPLETE WORKFLOW EXAMPLE

1. User places OCR file in monitoring directory
2. watcher.py detects the new file
3. db_ingestion chunks the text (1000 char chunks)
4. auto_extractor searches for covenant-related chunks
5. Chunks sent to Gemini LLM for metric extraction
6. Metrics saved to financial_metrics table
7. User opens Streamlit dashboard
8. Dashboard fetches latest metrics from database
9. risk_engine calculates breach probability via XGBoost
10. Dashboard displays metrics + risk gauge visualization

8. KEY TECHNOLOGIES

Technology	Purpose
Python 3.x	Core programming language
PostgreSQL (Neon)	Cloud database for documents & metrics
LangChain	Text chunking & splitting
Google Gemini API	LLM-based metric extraction
XGBoost	Machine learning risk model
Streamlit	Web dashboard UI framework
Plotly	Interactive data visualization
Watchdog	File system monitoring
psycopg2	PostgreSQL Python driver

9. PROJECT GOALS

- **Automate Extraction:** Extract financial covenants from quarterly/IM documents
- **Validate Compliance:** Validate compliance against regulatory limits
- **Predict Default Probability:** Use ML models to predict default/breach probability
- **Visualize Risk:** Display risk metrics through interactive dashboard

Key Financial Metrics Extracted:

Metric	Target Limit	Status	Significance
Capital Adequacy Ratio	15%	Higher is safer	Bank stability & regulatory compliance
Leverage Ratio	4.0x	Lower is safer	Debt burden relative to equity
Asset Quality (Net NPA)	1-2%	Lower is safer	Credit quality of loan portfolio
Security Cover	1.1x+	Higher is safer	Collateral coverage for bondholders

This document provides a comprehensive overview of the BondRisk AI project architecture, components, and workflow. For technical implementation details, refer to individual Python source files in the repository.